# Calibrate Where You Deploy: Conformal Sensor Escalation Under Operating-Point Shift ⋆

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**Abstract:** Condition-monitoring nodes keep a cheap sensor on and wake an expensive one on demand; a split-conformal gate replaces the tuned wake-up threshold, so risk becomes the specification. On KAIST and Paderborn the gate matches a threshold in accuracy, but its guarantee, which survives unseen bearings, collapses under operating-point shift: nominal 80 % delivers 33–70 %. With amplitude features, speed breaks every modality, torque mainly motor current. At $\alpha = 0.2$, five labelled 4-s recordings from the new point restore mean coverage at a higher escalation rate; pooling them with stale data is worse than replacing it. Recalibration must track the operating point.

*Keywords:* Condition and health monitoring; machine learning based FDI; statistical methods; data-driven diagnosis methods; predictive maintenance strategies; conformal prediction; distribution shift; sensor selection.

Table 1. Datasheet typical power of the sensing chains and of the MCU (ESP32-S3, 240 MHz, peripheral clocks off; other settings move the MCU rows by up to about 3×), and energy per 160 ms window. Last row: computation of the accelerometer time+FFT tier (71 264 cycles, 16-bit ESP-DSP FFT, extrapolated), on a marginal (543 pJ per cycle) / total-power basis.

| component | mW | mJ / window |
|---|---|---|
| IIS3DWB MEMS accel., 3 axes, on-chip ADC | 3.30 | 0.53 |
| LEM CKSR 15-NP transducer + ADC, per phase | 81.66 | 13.07 |
| TMCS1100 Hall sensor + ADC, per phase | 22.52 | 3.60 |
| MCU, MEMS window: light sleep + 10 FIFO drains | – | 2.60 |
| MCU, current window (no FIFO): idle | 108.57 | 17.37 |
| MCU compute, accel. time+FFT (marg. / total) | – | 0.039 / 0.071 |

## 1. INTRODUCTION

A battery-powered condition-monitoring node is dominated, energetically, by acquisition: keeping a sensing chain powered for a 160 ms window costs far more than classifying that window (Table 1). The usual answer is a cascade. A cheap *tier* (a set of channels and the features computed from them) runs on every window, and the expensive tier is woken only when the cheap tier looks uncertain: in supervision terms, a residual evaluation whose threshold schedules the acquisition of the expensive tier (Isermann, 2006). The weak point is the wake-up rule, in practice a *threshold gate*, $\max_y \hat{p}(y \mid x) < \tau$ on the top-class score of the cheap classifier, with $\tau$ swept by hand. Such scores are not calibrated probabilities: $\tau$ fixes a duty cycle (as a quantile of the calibration scores) but says nothing about the risk of a wrong decision, and current practice specifies neither.

Split conformal prediction (Vovk et al., 2005) returns prediction *sets* with finite-sample marginal coverage $\mathbb{P}(Y \in C(X)) \geq 1-\alpha$ under exchangeability and with no assumption on the classifier. Using the set as the gate (escalate iff $|C(x)| \neq 1$) replaces $\tau$ by a stated risk $\alpha$ and makes the escalation rate a measurement. This rule has been applied to cascades of *models*, pruning candidate labels stage by stage (Fisch et al., 2021) or routing images and queries between light and heavy tiers by prediction-set size (Turkoglu et al., 2026; Dou et al., 2026); there the cascaded object is a model and the currency is computation. Here the object is a *sensor* and the currency is acquisition energy.

Two questions decide whether such a gate can be trusted on a machine. First, the guarantee is conditional on exchangeability between calibration and deployment data, which a change of operating point violates (Tibshirani et al., 2019; Barber et al., 2023). Conformal thresholds are in use for false-alarm control in process monitoring (Diallo et al., 2025); regime-conditional under-coverage has been reported for remaining-useful-life regression (Yang et al., 2026); and cross-domain sets for bearing diagnosis reach nominal coverage with calibration drawn from the target domain (Mei et al., 2025). Static conformal thresholds collapse across simulated process-mode transitions (Heddoub et al., 2026b); the measurement has not been made for sensor-based fault classification on physical benchmarks. Second, what does restoring it cost? Label-complexity

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results indicate that under unrestricted shift only target labels restore coverage (Han et al., 2026); no commissioning number exists for condition monitoring.

This paper answers both questions on two public benchmarks with one pipeline: the guarantee does not survive a change of operating point (80 % requested, 33–70 % delivered) although it survives unseen bearings, and five labelled recordings at the new point restore it on average. (i) The set-size gate is moved from model cascades to sensor wake-up and priced in datasheet energy; the only cascade that saves acquisition energy, current always on and vibration on demand, is the one most exposed to shift. (ii) At matched escalation fraction the conformal and the threshold gate are indistinguishable in accuracy, both in distribution and under shift; the gate's value is the specification, not the predictor. (iii) With source-only calibration, nominal 80 % falls to 40.2–70.3 % (KAIST) and 32.6–50.0 % (Paderborn) by classifier, in one of two failure modes that held-out accuracy does not predict; on Paderborn speed breaks every tier, torque mainly the current tier. (iv) Five labelled recordings from the new operating point restore mean coverage to within 0.01 of nominal, with a per-deployment spread of about ±0.15 and at a higher escalation rate; pooling them with the stale source scores, weighted or not, helps only in proportion to the target share of the calibration weight.

## 2. CONFORMAL SENSOR ESCALATION

### *2.1 Split conformal prediction*

Let $\hat{p}(\cdot \mid x)$ be any classifier over $K$ classes and $\{(x_i, y_i)\}_{i=1}^{n}$ a calibration set disjoint from training. With the least ambiguous set-valued classifier (LAC) score $s(x, y) = 1 - \hat{p}(y \mid x)$ (Sadinle et al., 2019) and $\hat{q}$ the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration score, the prediction set is

$$C(x) = \{\, y : s(x, y) \le \hat{q} \,\}. \tag{1}$$

If calibration and test points are exchangeable, $\mathbb{P}(Y \in C(X)) \ge 1 - \alpha$ for any finite $n$ (Vovk et al., 2005). Under weighted conformal prediction (Tibshirani et al., 2019; Barber et al., 2023) calibration score $i$ carries a weight $w_i \ge 0$, the test point weight 1, and $\hat{q}$ is the $(1 - \alpha)$-quantile of the weighted empirical distribution normalised by $\sum_i w_i + 1$; with $w_i \equiv 1$ this is (1). The windows of one recording are dependent, so calibration scores are exchangeable at recording level, not window level; the finite-sample statement then holds approximately, with an effective calibration size equal to the number $k$ of recordings (Dunn et al., 2023): exact recording-level validity needs $\alpha \ge 1/(k+1)$, and the realised coverage of one calibration draw spreads around $1-\alpha$ with a standard deviation of order $\sqrt{\alpha(1-\alpha)/k}$ (Vovk, 2012).

### *2.2 The gate*

The cheap tier runs on every window; the expensive tier is woken, and its classifier decides, exactly when the cheap tier's set is not a singleton:

$$\text{escalate}(x) \iff |C_{\text{cheap}}(x)| \ne 1. \tag{2}$$

Anything but a singleton (several classes, or none) is an admission that, at risk $\alpha$, the cheap sensor cannot resolve the window. Because (1) is marginal, the gate does not promise that the singleton is correct; it promises that the fraction of all windows on which the node emits a wrong singleton is at most $\alpha$, and it turns the escalation fraction, hence the duty cycle of the expensive sensor, into an observable. With the LAC score, $|C(x)| = 1$ iff the top-class probability is at least $1 - \hat{q}$ and the second is below it; when $1 - \hat{q} > 0.5$ this is a threshold on the top-class score, which is why the threshold gate is the right baseline, but in the experiments $\hat{q}$ is large ($1 - \hat{q} > 0.5$ only for the Paderborn random forest from $\alpha = 0.15$–$0.20$ upwards), so the two gates are different selection rules and their comparison in Sec. 4.2 is empirical. In model-based terms the operating point is a scheduling variable the classifier does not see; conditioning the calibration on its tag (Mondrian conformal prediction, Vovk et al., 2005) is the data-driven counterpart of a scheduled residual threshold (Isermann, 2006).

### *2.3 Cost: operations and energy*

Operations per window: $9N$ per channel for the nine time-domain statistics, plus $\approx 2.5N \log_2 N + 2N$ with the FFT block ($N = 4096$). Energy is modelled from datasheet typicals (Table 1): the chain of a tier is powered for the 160 ms window, and arithmetic is priced in picojoules per cycle. An active current transducer draws, per phase, 7–25× the power of a three-axis digital micro-electromechanical-systems (MEMS) accelerometer, so with dedicated active sensing the "cheap current" tier is the expensive one in joules. The MCU is not free during acquisition either: the MEMS FIFO holds about a ninth of a window, so the MCU sleeps between ten FIFO drains over SPI and spends 2.6 mJ, two to seven times the sensor energy depending on clock setting; the current ADC has no FIFO, so on a dedicated current tier the ESP32-S3 idles for the whole window (17.4 mJ). The computation of the accelerometer time+FFT tier is 0.071 mJ per window with the 16-bit FFT kernel and several times that with a float FFT and the descriptor arithmetic, five to forty times below the 2–4 mJ of acquiring it (sensor plus MCU service), so a same-sensor cascade saves operations (4.6×), not acquisition energy. The only cascade that saves acquisition energy, current always on and vibration on demand, requires current to be free: the node is the drive controller that already samples phase current (at its own rate and resolution, on which the current-tier classifier must then be trained) and the accelerometer its wired or wireless satellite (radio wake-up not modelled); the cascade's energy per window is then the MEMS acquisition energy times the escalation fraction. A MEMS window with its MCU service, 3.1 mJ, equals the light-sleep floor (0.79 mW) at one window every 4 s, so the joules matter for near-continuous monitoring.

## 3. EXPERIMENTAL SETUP

**Data.** KAIST (Jung et al., 2023): a motor-driven shaft (induction motor, constant 3010 rpm) with four accelerometers and three current transformers (Hioki CT6700), vibration and current both at 25.6 kHz; five conditions at three torque loads (0, 2, 4 N m): normal (one recording per load), inner- and outer-race faults, misalignment and unbalance at three to five severities each, pooled per class; 45 recordings; only the U-phase current is present in every

Table 2. Cost–accuracy frontier: macro-F1 of the better of LR and RF per tier (mean over three seeds for the grouped protocols; one run per held-out point under shift) with 95 % percentile intervals of a cluster bootstrap over recordings (bearings for the bearing-grouped column) of the seed-0 predictions. Operations per 160 ms window.

| | tier | ops / window | recording-grouped | bearing-grouped | operating-point shift |
|---|---|---|---|---|---|
| KAIST | current, time+FFT | 167 936 | 0.403 [0.32, 0.46] | – | 0.213 [0.13, 0.29] |
| | 1 accel., time | 36 864 | 0.690 [0.66, 0.76] | – | 0.687 [0.63, 0.74] |
| | 4 accel., time+FFT | 671 744 | 0.729 [0.70, 0.91] | – | 0.746 [0.68, 0.81] |
| Paderborn | 2 current phases, time+FFT | 335 872 | 0.905 [0.88, 0.92] | 0.458 [0.34, 0.61] | 0.612 [0.58, 0.64] |
| | 1 accel., time | 36 864 | 0.887 [0.87, 0.91] | 0.548 [0.41, 0.68] | 0.714 [0.68, 0.75] |
| | 1 accel., time+FFT | 167 936 | 0.959 [0.95, 0.97] | 0.553 [0.40, 0.67] | 0.754 [0.72, 0.79] |
| | 1 accel. + 2 phases | 503 808 | 0.985 [0.98, 0.99] | 0.504 [0.37, 0.63] | 0.768 [0.74, 0.80] |

file. Paderborn (Lessmeier et al., 2016): 32 bearings (6 healthy, 12 with machined damage, 14 with damage from accelerated-lifetime tests), one vibration channel and two current phases at 64 kHz, four operating points, three of which differ from the baseline (1500 rpm, 0.7 N m, 1000 N) in exactly one factor: speed 900 rpm, torque 0.1 N m, radial force 400 N; four classes (healthy, outer race, inner race, combined: three bearings, all real damage). Signals are resampled to 25.6 kHz with an exact 2/5 polyphase filter, which discards the band above 12.8 kHz in which envelope analysis would demodulate bearing resonances (Randall and Antoni, 2011); five of the twenty measurements per bearing and operating point are used (639 of 640 recordings, one KA08 file unreadable; 31 266 windows).

**Features and tiers.** Windows of 4096 samples with hop 2048, at most 50 per recording (about 4 s); per channel nine time-domain statistics (mean, standard deviation, RMS, peak-to-peak, kurtosis, skewness, crest, shape and impulse factors) and, when the FFT block is included, six spectral descriptors (dominant frequency, centroid, spectral entropy, energy in 0–2, 2–6 and 6–12.8 kHz).

**Classifiers.** $\ell_2$-regularised logistic regression (LR, $C = 1$ on standardised features) and a random forest (RF, 300 trees, minimum leaf 2); convergence is checked for every fit. A multilayer perceptron (64 and 32 hidden units) was also run; on the tiers of Table 2 it never beats the better of LR and RF by more than 0.014 macro-F1 and is omitted.

**Validation.** All splits are at recording level. *Recording-grouped* stratified 3-fold cross-validation (three seeds) estimates in-distribution performance ("in" in the tables); *operating-point shift* trains on all but one operating point and tests on it. On Paderborn the same bearing is recorded at every operating point, so recording-grouped folds let a bearing appear on both sides; a *bearing-grouped* protocol (folds and calibration sets carved by bearing) bounds that optimism. The shift protocol shares bearings by construction, so the degradation it exposes is due to the operating point alone. Intervals are 95 % percentile intervals of a cluster bootstrap over recordings (600 replicates).

**Conformal protocol.** A stratified third of the training recordings is held out for calibration in every fold; the calibration scores of the folds are pooled and one $\hat{q}$ is applied to the pooled test predictions (a cross-conformal construction, Vovk, 2015; the recalibration protocol below, one quantile per point, carries the exact guarantee), so that under shift the pooled calibration set holds scores from all source points. Coverage is the fraction of test windows whose label lies in $C(x)$, and the *gap* is coverage minus $1 - \alpha$, for $\alpha \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$. The conformal gate is compared with the threshold gate at matched escalation fraction by interpolating the threshold curve swept on the test windows themselves, so the baseline is an oracle in $\tau$ and the comparison cannot favour the conformal gate. For recalibration each held-out operating point is the target: the classifier is trained on the others and the calibration set is the source recordings only, the source recordings pooled with $k$ labelled target recordings, or the $k$ target recordings alone, with $k \in \{1, 2, 3, 5, 8\}$ drawn at random (unstratified) over five seeds and $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$; test recordings are disjoint from calibration recordings. Weighted conformal prediction gives the source scores a fixed weight $\rho \in \{1, 0.3, 0.1, 0.03, 0\}$ and the target scores weight 1.

## 4. RESULTS

### *4.1 How much sensing is needed*

Two cascades are evaluated, each with the full vibration tier of its dataset as the expensive tier (four accelerometers on KAIST, one on Paderborn, with FFT): current-first (*current → vibration, time+FFT*), the only one that can save acquisition energy, and same-sensor (*one accelerometer, time-domain → vibration, time+FFT*), which saves computation. On KAIST (Table 2), nine time-domain features from a single accelerometer reach 0.69 macro-F1 in distribution and under shift, against 0.73/0.75 for four accelerometers with FFT at 18× the operations, and current alone collapses under shift (0.40 to 0.21) while escalating 83–100 % of windows (seed means) in distribution at every $\alpha \leq 0.20$, so the KAIST current-first cascade is studied for its guarantee, not its saving. On Paderborn, two current phases reach 0.91 under recording-grouped folds, near vibration, but the bearing-grouped column shows that recording-grouped folds recognise known bearings rather than damage on unseen ones: every tier falls to 0.46–0.55. Under operating-point shift, vibration degrades (0.959 to 0.754) and current collapses (0.905 to 0.612).

### *4.2 The gate does not buy accuracy*

The conformal gate is compared with the oracle threshold gate at matched escalation fraction, i.e. at equal cost. In distribution, pooled over cascades, classifiers and seeds, the macro-F1 difference, conformal minus threshold (mean ± s.d. over matched points with 2–98 % escalation), is $-0.0008 \pm 0.0051$ on KAIST and $-0.0028 \pm 0.0052$ on Paderborn, and within $\pm 0.004$ for every cascade; under

Table 3. Coverage gap (empirical − nominal) of LAC sets for the current tier (current, time+FFT on KAIST; two phases on Paderborn): mean and worst cell over the $\alpha$ grid, classifiers and seeds (48 cells per row); coverage at nominal 80 % by LR / RF, mean ± s.d. over seeds, and the same averaged over classifiers for the healthy and the fault classes. in = recording-grouped, bearing = bearing-grouped, shift = operating-point shift.

| dataset | protocol | mean | worst | LR / RF at 80 % | healthy / faulty |
|---|---|---|---|---|---|
| KAIST | in | +0.045 | −0.019 | 82.9±2.1 / 93.5±11.2 | 38 / 92 |
| KAIST | shift | −0.148 | −0.453 | 40.2±5.4 / 70.3±6.4 | 33 / 56 |
| Paderborn | in | −0.009 | −0.039 | 79.0±1.3 / 78.2±0.9 | 69 / 82 |
| Paderborn | bearing | +0.002 | −0.053 | 81.2±4.7 / 80.8±1.0 | 62 / 82 |
| Paderborn | shift | −0.296 | −0.501 | 50.0±5.9 / 32.6±2.6 | 57 / 37 |

shift it is $-0.0037 \pm 0.0145$ and $-0.0016 \pm 0.0048$. This negative result is the point: the gate is not a better predictor; it lets $\alpha$ fix the duty cycle without tuning.

### *4.3 Coverage collapses under operating-point shift*

Table 3 and Fig. 1(a) give the central result for the current tier, Table 4 the vibration tiers. Under recording-grouped validation the marginal guarantee holds (mean gap +0.045 on KAIST, −0.009 on Paderborn), and under bearing-grouped validation as well (+0.002 mean, worst −0.053): marginal coverage does not depend on the accuracy of the classifier. Under operating-point shift it fails: mean gap −0.148 (worst cell −0.453; −0.743 at a single held-out point) on KAIST and −0.296 (−0.501; −0.825) on Paderborn; at nominal 80 % the gate delivers 40.2 % LR / 70.3 % RF on KAIST and 50.0 % LR / 32.6 % RF on Paderborn, and the shift curves of Fig. 1(a) lie below the diagonal at every $\alpha \geq 0.10$: $\alpha$ moves the coverage but no longer sets it. The bearing and shift rows differ in what they break: bearing-grouped folds cut the macro-F1 of the current tier from 0.91 to 0.46 (Table 2) and leave the gap at +0.002; the operating-point shift costs less accuracy (0.91 to 0.61) and 0.30 of coverage; held-out accuracy is no proxy for the validity of the gate. In the currency of (2), the fraction of windows with a wrong singleton, LR promises 0.20 at $\alpha = 0.20$ and delivers 0.44 (KAIST) and 0.46 (Paderborn) under shift; RF stays within that bound (0.06, 0.05) by escalating instead, but not at $\alpha \leq 0.10$ on Paderborn (0.20 against 0.10); in distribution every cascade meets it at $\alpha \leq 0.10$. The healthy/faulty column is the diagnostic reading: marginal coverage says nothing about the healthy class, which on the KAIST current tier is covered at 38 % in distribution (one healthy recording per load; the RF only through full label sets) while the fault classes are covered at 92 %; a false-alarm bound needs class-conditional calibration (Heddoub et al., 2026a).

How the gate fails depends on how the cheap classifier behaves off distribution (Table 4, right), and only one of the two modes is silent. Logistic regression keeps emitting singletons under shift (0.62–0.96 of the sets) while coverage falls to 0.46–0.68: confident *and wrong*, so the gate under-escalates. The random forest instead returns empty sets (0.47–0.75): a quantile calibrated on confident in-distribution probabilities is small, under shift no class clears it, and by (2) the gate over-escalates, wasteful but safe (only because (1) admits the empty set); which mode

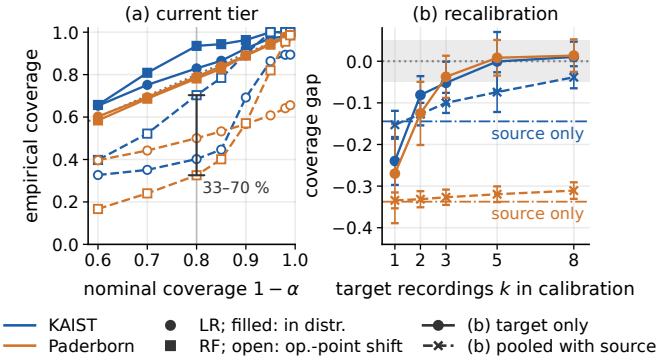



Fig. 1. (a) Empirical against nominal coverage of the LAC gate on the current tier, in distribution (filled) and under operating-point shift (open), mean over seeds; dotted diagonal: the guarantee. (b) Coverage gap against the number $k$ of labelled target recordings, calibrated on them alone or pooled with the source scores, mean over classifiers, $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ and held-out points, ± s.d. of that mean over five draws; dash-dotted: source-only calibration; band: ±0.05.

occurs depends on classifier and calibration set (the same-sensor RF gate below is silent). The silent mode's direction is modality-specific and physical: at 900 rpm every bearing is quieter, so the LR vibration gates under a speed change still cover the healthy class fully and the fault classes at 0.03–0.55 (missed detections), whereas the LR current classifier under a speed or torque change emits one fault class for every window and covers the healthy class at 0.00 (false alarms). On Paderborn, in the same-sensor cascade the RF gate at $\alpha = 0.10$ escalates 6 % of windows in distribution and 6 % under shift while macro-F1 falls from 0.890 to 0.719 and the wrong-singleton fraction reaches 0.28; in the current-first cascade with free current, $\alpha = 0.05$ with RF escalates 33 % in distribution (1.0 mJ of vibration acquisition per window instead of 3.1 mJ) for macro-F1 0.939, and 53 % under shift (26–70 % across calibration draws) for 0.722. An operating-point change moves the node along the energy–accuracy curve with no change in $\alpha$; the escalation fraction, its only signal, moves in three of these four cases and not in the first, so it can trigger a recalibration but cannot certify its absence.

### *4.4 Which factor breaks exchangeability, per modality*

Three Paderborn operating points each change one factor, so classifiers fitted and calibrated at the baseline can be evaluated on each neighbour (Table 4, left). For the vibration tiers a change of speed destroys coverage (0.07–0.50), a change of torque leaves it near nominal for LR (0.80–0.82) and 0.09–0.13 below nominal for RF, and a change of radial force degrades it moderately for LR (0.64–0.73) and severely for RF (0.27–0.40); for the current tier both speed and torque destroy it (0.01–0.38). Vibration-only domain-shift studies on this benchmark likewise find speed dominant (Panić et al., 2027); the torque sensitivity belongs to the current tier, as the KAIST (torque) shift column confirms (vibration 0.69 to 0.69, current 0.40 to 0.21). The feature statistics agree: at 900 rpm the vibration RMS of healthy bearings falls by a quarter and that of damaged ones by half, so the amplitude statistics move and

Table 4. Paderborn, LAC at nominal 80 %, classifiers fitted and calibrated at the baseline operating point: coverage (macro-F1) on held-out baseline recordings (in; recording-grouped, so the macro-F1 reference is optimistic) and on each point that changes one factor; coverage, singleton and empty-set fractions over the three shifted points; seed s.d. $\leq 0.12$.

| | | coverage (macro-F1) by factor changed | | | | shift, pooled | | |
|---|---|---|---|---|---|---|---|---|
| tier | classifier | in | speed | torque | radial force | coverage | singletons | empty |
| 2 current phases, time+FFT | LR | 0.77 (0.74) | 0.38 (0.14) | 0.35 (0.13) | 0.66 (0.57) | 0.46 | 0.96 | 0.00 |
| | RF | 0.78 (0.92) | 0.01 (0.26) | 0.15 (0.43) | 0.46 (0.66) | 0.21 | 0.25 | 0.75 |
| 1 accel., time | LR | 0.82 (0.66) | 0.50 (0.31) | 0.82 (0.64) | 0.73 (0.56) | 0.68 | 0.62 | 0.00 |
| | RF | 0.79 (0.93) | 0.08 (0.22) | 0.71 (0.86) | 0.40 (0.59) | 0.40 | 0.53 | 0.47 |
| 1 accel., time+FFT | LR | 0.83 (0.80) | 0.36 (0.27) | 0.80 (0.76) | 0.64 (0.63) | 0.60 | 0.91 | 0.01 |
| | RF | 0.86 (0.98) | 0.07 (0.26) | 0.67 (0.90) | 0.27 (0.62) | 0.34 | 0.39 | 0.61 |

the scale-free ones (kurtosis, crest factor) move with the number of impacts per window; the phase-current RMS falls by only 4 %, but that is 2.6 within-point standard deviations, more than the fault information carries. Torque from 0.7 to 0.1 N m halves the phase-current RMS (1.76 to 0.91 A): every amplitude statistic of the current channel rescales while the fault content does not, the classical load dependence of motor-current signature analysis, which the gate inherits; vibration RMS is unchanged. Radial force (1000 to 400 N) moves the current statistics by under 1 % and the vibration RMS by 6–13 %, enough for the sharp RF quantile of Sec. 4.3 to empty its sets. One step per factor, of unequal size, is what the benchmark offers. The design consequence, for the amplitude-based, fixed-band features used here: a current-first cascade must be recalibrated whenever speed *or* torque changes; a vibration tier tolerates torque changes, degrades moderately (LR) to severely (RF) under radial force, and needs recalibration for speed.

### *4.5 What it takes to fix it*

Fig. 1(b) quantifies recalibration, i.e. Mondrian calibration on the operating-point tag (Sec. 2.2), against the number $k$ of recordings available from the target operating point. Five recordings (250 windows, about 20 s) restore mean coverage to within 0.01 of nominal on average over the $\alpha$ grid (gap from $-0.144$ to $-0.001$ on KAIST and from $-0.337$ to $+0.008$ on Paderborn) and three to within about 0.05 ($-0.052$, $-0.037$). That is a mean: for one deployment the realised coverage at $k = 5$ spreads with s.d. 0.16 (KAIST) and 0.13 (Paderborn) over classifiers, points and draws, and 37 % / 28 % of the configurations stay more than 0.05 under nominal; five at $\alpha = 0.20$ is the recording-level minimum $\lceil (1-\alpha)/\alpha \rceil = 4$ plus one, a statement about $\alpha$ before it is one about either rig (at $\alpha = 0.05$ the rule asks for 20 and the $k = 5$ gap is still $-0.03$ on Paderborn). The number is in *labelled* recordings: cheap on a rig or in a fleet; a production machine has only healthy recordings at the new point, which can recalibrate the healthy-class quantile alone (untested here). Restoring the guarantee has a second price, in escalations: the classifier is still the one fitted at the source points, so coverage returns only through larger sets. At $\alpha = 0.10$ the recalibrated RF current gate on Paderborn escalates 68 % of windows against 12 % in distribution (LR 91 % against 68 %); over the $\alpha$ grid the current tier escalates 92 % (KAIST) and 72 % (Paderborn) after recalibration against 75 % and 42 % with the stale quantile. Recalibration makes the gate honest at the new point, not cheap (retraining untested).

Pooling the target recordings with the stale source set is worse than replacing it: at $k = 3$ the target-only gap on KAIST is $-0.052$ against $-0.100$ pooled; on Paderborn the source set holds about 7 729 windows against about 50 per target recording, so pooling barely moves the quantile ($-0.327$ at $k = 3$, $-0.311$ at $k = 8$) while target-only reaches $-0.037$. This is arithmetic: with target share $\pi = n_{\mathrm{tgt}}/(n_{\mathrm{tgt}} + \rho\, n_{\mathrm{src}})$ of the calibration weight ($\rho = 1$ for pooling, $\rho < 1$ for weighted conformal prediction with source weight $\rho$ and target weight 1), the quantile is that of the mixture $(1-\pi)F_{\mathrm{src}} + \pi F_{\mathrm{tgt}}$, so target coverage is at least $1 - \alpha/\pi$, a bound that is vacuous once $\pi \leq \alpha$. On Paderborn pooling has $\pi \leq 0.05$; $\rho = 0.3$ at $k = 5$ still leaves the source in control ($-0.271$ against $+0.008$ target-only) and only $\rho = 0.03$ ($\pi \approx 0.5$) approaches it ($-0.056$).

### *4.6 Limits of recalibration*

Recalibration restores the guarantee only while the classifier is informative: training on machined damage and testing on real damage (and vice versa) gives macro-F1 of 0.11–0.35 for every tier and classifier (a three-class model on four classes, capped at 0.75): chance. Target-only recalibration then restores coverage only by inflating the sets, so the gate would escalate every window. Operating-point shift is a calibration problem that a few labelled recordings fix for validity, at the escalation fraction the degraded classifier deserves; damage-origin shift is a representation problem they do not fix.

## 5. DISCUSSION

A conformal escalation gate is a better *interface* than a threshold gate, not a better predictor, and its guarantee expires when the machine leaves the operating point it was calibrated at. Three commissioning steps follow. (1) Calibrate at the operating point of deployment. (2) When the operating point changes (speed for any tier; torque as well for a current tier; radial load for a random-forest vibration tier), discard the stale scores and recalibrate on $k \geq \lceil (1-\alpha)/\alpha \rceil + 1$ labelled recordings with faults present (verified for $k \leq 8$), expecting a spread of order $\pm\alpha$ for a single commissioning and a higher escalation rate; a variable-speed drive needs per-speed (Mondrian) calibration or speed-normalised features, neither tested here. (3) Take the trigger from outside the node (the drive's operating-point tag or a drift monitor): neither failure mode is certifiable from its outputs; a moving escalation rate warns; a constant one is no all-clear.

**Limitations.** Energy figures are datasheet typicals, duty-cycled per window, with turn-on, sleep transitions and radio excluded; the priced IIS3DWB (6.3 kHz passband) misses the top band of Sec. 3; the compute figures are extrapolated from published kernel cycle counts. Both benchmarks use hand-crafted features and shallow classifiers; the coverage statements are classifier-agnostic, but the transfer failures may not be. Only labelled target data and fixed source weights are tested; label-free recalibration from unlabelled target windows (Yilmaz and Heckel, 2022) and feature-space likelihood-ratio weighting remain to be compared.

**Code and data.** Both datasets are public; code, executed notebooks and results are at commit 7828be2 of `https://github.com/candidoalfilho/conformal_escalation`.

## 6. CONCLUSION

Sensor escalation is formulated as a conformal set-size test: a stated risk replaces a tuned threshold and the escalation rate becomes a measurement; on two benchmarks the gate matches a threshold in accuracy, and its coverage guarantee, valid in distribution and on unseen bearings, fails under operating-point shift, delivering 33–70 % (two datasets, two classifiers) where 80 % was requested, a failure that held-out accuracy does not predict. Five labelled recordings from the new operating point restore mean coverage, with a per-deployment spread of about $\pm 0.15$ and at a higher escalation rate; pooling them with the stale calibration set helps only in proportion to their share of the calibration weight. Conformal calibration is a commissioning step tied to the operating point.

## DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors used Claude (Anthropic) in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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